The Soft-Lock Algorithm: A Technical Description

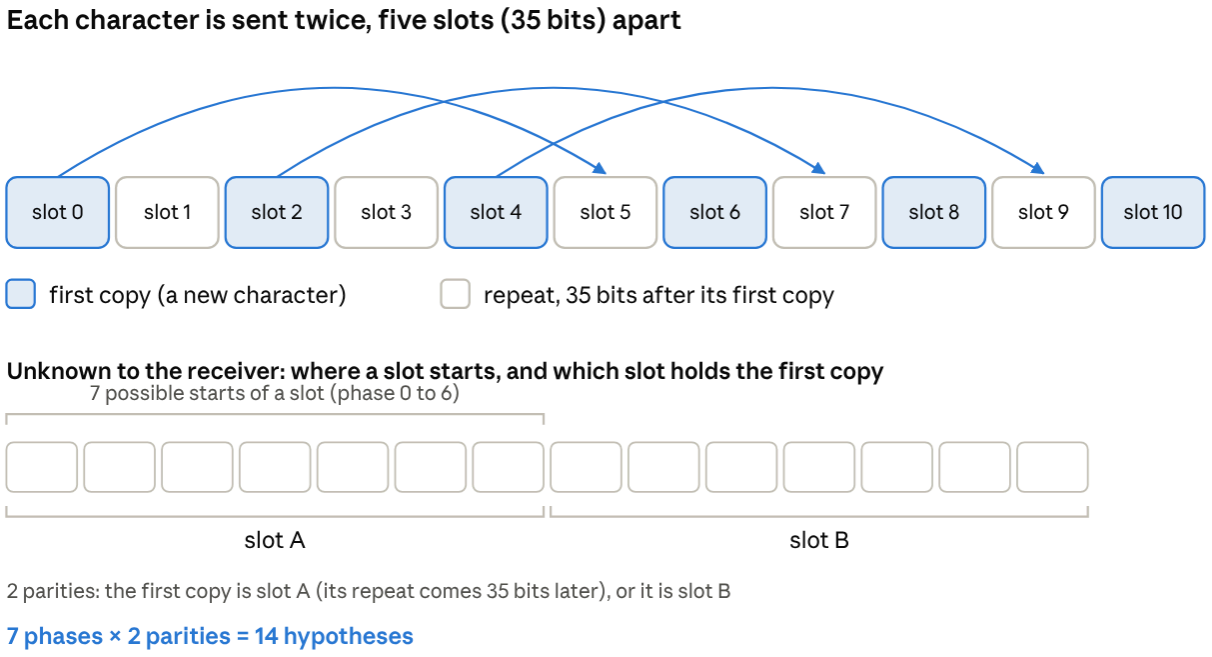
Nov 6, 2026 · @brian martlew

The soft-lock recovers SITOR-B text from signals too weak for hard bit decisions, by scoring 14 candidate framings of the bit stream on how well each shows the mode's built-in 35-bit repetition, accumulating that evidence over a window, and decoding only after one framing has clearly won.

Signal structure

SITOR-B sends 100 bits/s in 7-bit codes. Every valid character has exactly four 1-bits, so one wrong bit shows up as a weight of three or five. Each character is sent twice: the repeat follows five 7-bit slots (35 bits, 350 ms) after the first copy, and the slots alternate between first copies and repeats.

The stream carries no frame marker, so the receiver must find two unknowns. The **phase** is which of 7 bit positions starts a slot. The **parity** is whether the first copy sits in the even or the odd slot of each pair. That gives 7 × 2 = 14 hypotheses. Everything below is a way of choosing among them from soft values, without first committing to hard bit decisions.

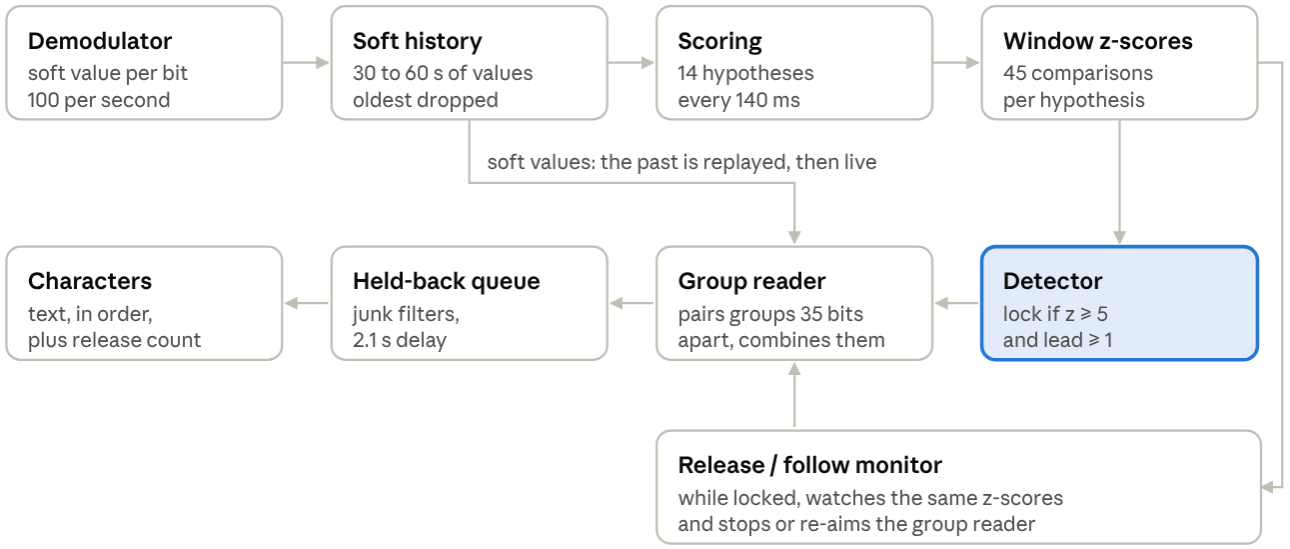


repeat structure and the 14 hypotheses

Overview

The demodulator delivers one soft value per bit: the sign is the likely bit, the magnitude is the confidence. The lock stores these in a history of 30 to 60 s, because it decides late and then applies the decision to the past. While searching it only scores and accumulates. Once it locks, a group reader decodes from the history and then follows the live stream, and the same scores are watched for signs that the signal has gone or changed.

Scoring finds the framing; the group reader then decodes from the history



processing pipeline, searching and locked

The top row runs all the time. The bottom row, read right to left, runs only after a lock.

Scoring a hypothesis

Every 140 ms (two slots, 14 bits) each hypothesis forms one pair of groups: **x**, seven soft values at the hypothesised slot, and **y**, the seven values 35 bits later. If the framing is right, they are two noisy copies of one character. The score adds two measures:

$$s = \frac{c - \mu_c}{\sigma_c} + \frac{q - \mu_q}{\sigma_q}, \qquad c = \frac{x \cdot y}{|x|\,|y|}, \qquad q = \max_{k} \frac{(x+y) \cdot k}{\sqrt{7}\,|x+y|}$$

- **c, agreement.** The cosine of the angle between the two groups. It is near +1 when the copies agree in sign and relative size, and near 0 for unrelated data.
- **q, validity.** The groups are added bit by bit, so noise partly cancels and signal reinforces. The sum is correlated with each of the 32 traffic codes (patterns of ±1, with four +1s) and the best match is kept.

Each term is shifted and scaled by its mean and standard deviation on pure noise (c: 0.003 and 0.379; q: 0.715 and 0.128). A wrong framing therefore scores about 0 with a standard deviation of 1.42, while the right one scores a little above 0 every time. A single score is almost useless, which is why the evidence is accumulated.

Detecting the lock

The last 45 scores of each hypothesis (6.3 s) are summed and divided by the noise-only standard deviation of such a sum:

```
z_h = \frac{\sum_{i=1}^{45} s_{h,i}}{\sigma \sqrt{45}}, \qquad \sigma = 1.4168
```

On noise, z is roughly standard normal. The detector locks to the best hypothesis when its z is at least 5 and it leads the runner-up by at least 1. The lead rule requires the winner to stand clear of the field, not merely to cross the threshold.

The threshold was set by measuring false locks on recorded noise. At z = 5 there were none in 24 hours; z = 4 gave about 0.3 an hour and z = 3.5 about 1.8 an hour. It is tied to the window length and the noise statistics, so none of the three should be changed alone.

Detection takes several seconds, but no text is lost. When the lock is declared, decoding restarts from the first slot boundary of the winning phase about 7.5 s earlier (the window plus a margin of 10 comparisons), or from the end of the previous lock if that is later. The history makes this replay possible.

Decoding with soft combining

Once locked, the group reader steps through the history in 7-bit groups. Each repeat group is paired with the group five slots before it, and the pair goes to the combiner. The first five groups after a lock have no partner yet and are skipped.

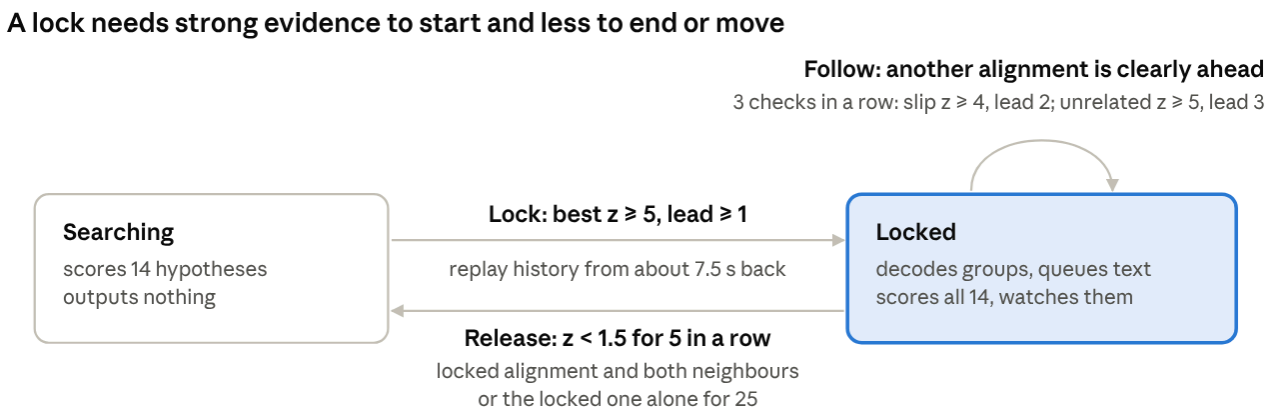
The combiner adds the two copies' soft values bit by bit and decides once, so a bit that one copy lost to noise is often rescued by the other. Then:

1. Four 1-bits in the sum: a valid code, accepted.
2. Three or five: one bit from valid. The least certain bit is flipped.
3. Otherwise the copies are judged separately. If one is valid, it is used. If both are valid but differ, the output is `!`. If neither is, the output is `~`.

Pairs of phasing signals print nothing. The combiner also tracks the letters/figures shift state, and this is reset after a jump to an unrelated alignment, since the shift state cannot be trusted across it.

Tracking: release and follow

After a lock, all 14 hypotheses keep being scored. The same scores drive two decisions, and the thresholds differ from the detector's because ending or moving a lock needs less evidence than creating one.



lock states and transitions

Release. The z-scores are taken over the last 30 comparisons. If the locked hypothesis and both of its one-bit-slip neighbours stay below z = 1.5 for 5 comparisons in a row (0.7 s), the signal is judged to have gone. Including the neighbours stops a one-bit slip of the bit clock from being mistaken for loss of signal. A last-resort rule releases if the locked hypothesis alone stays below 1.5 for 25 comparisons (3.5 s). On release the held-back characters are discarded as stray tail, the scores are cleared and a release counter is incremented, so the caller can start a new line.

Follow. Over the last 12 comparisons (1.7 s) the best hypothesis is compared with the locked one. A challenger must be clearly ahead on 3 successive checks. A one-bit slip needs z ⩾ 4 and a lead of 2. An unrelated alignment, such as a second, stronger station, needs z ⩾ 5 and a lead of 3.

Where did the change happen? The stored old and new score columns are searched for the split point that fits best: the old alignment before it, the new one after it.

```
k^{*} = \arg\max_{k} \left( \sum_{i \le k} s^{\mathrm{old}}_{i} + \sum_{i > k} s^{\mathrm{new}}_{i} \right)
```

Characters still held back from after the split are dropped, and the groups are re-read from the split under the new alignment.

Junk filters and the output delay

A phasing (idle) signal read out of step can produce plausible characters. Two filters act on the held-back queue, which is why characters wait about 2 s before output:

- **Letter runs.** Eight or more identical letters in a row are dropped, including the earlier letters of the run that are still queued.
- **Idle look-alikes.** Both groups of a pair are within one bit of the two phasing codes rotated by 1 to 6 places. The two copies of a real character agree, so real text cannot look like this. Three such pairs in a row are dropped, with the first two still queued. A genuine idle pair is left alone.

Parameters and performance

These are internal constants, not user settings. They were calibrated together with the window length and the noise statistics.

Parameter	Value	Role
Detection window	45 comparisons (6.3 s)	Evidence summed for a lock
Lock threshold	$z \geq 5.0$, lead ≥ 1.0	Minimum z and margin over the runner-up
Replay margin	10 comparisons	Decoding starts this far before the window
Output delay	15 comparisons (2.1 s)	Time characters are held back
Release	$z < 1.5$, 5 in a row	Signal judged gone (30-comparison window)
Last-resort release	25 in a row	Locked alignment alone is low
Follow window	12 comparisons (1.7 s)	Period over which challengers are compared
Slip challenger	$z \geq 4$, lead 2	One-bit neighbour takes over
Unrelated challenger	$z \geq 5$, lead 3	Different alignment takes over
Confirmations	3 checks	A challenger must stay ahead this long
Letter-run filter	8 letters	Identical letters in a row
Idle look-alike filter	3 pairs	Consecutive look-alike pairs

In the project's synthetic tests, 10% of characters were recovered at about −13 dB signal-to-noise with the soft-lock, against about −8 dB for the standard decoder. On real overnight logs from 490 and 518 kHz the gain is large. Some garbage remains near weak signals and where stations overlap, and this is accepted.

Note: dB figures are signal-to-noise ratios, with the noise measured in a 2,500 Hz bandwidth.